

Action Recognition from Photos

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Abstract

This project studies action recognition from one photo. The task is to classify an image as sitting, standing, or waving. We collected a new dataset from Wikimedia Commons instead of using Kaggle or another ready-made dataset. The final dataset has 360 images, with 120 images in each class. We compared a simple baseline based on HOG features and Logistic Regression with a transfer-learning model based on MobileNetV3-Small. On the fixed test set, the baseline reached 38.9% accuracy and the transfer model reached 77.8%. The result shows that pretrained CNN features are much more useful for this varied image dataset.

Keywords: action recognition, image classification, transfer learning, MobileNetV3, HOG

Introduction

Recognizing human actions is useful in photo organization, accessibility tools, human-computer interaction, and scene understanding. It is harder than simple object recognition because the important information is often the position of the whole body or a small part such as a raised hand. A single photo also does not show motion over time, so context and body pose become important.

The goal of this assignment was to complete the full computer vision process: collect data, clean and organize it, apply preprocessing and augmentation, train two models, and evaluate them with accuracy, precision, recall, F1-score, and confusion matrices. The same train, validation, and test split was used for a fair comparison.

Project summary

Item	Value
Classes	sitting, standing, waving
Images	360 total, 120 per class
Split	252 train, 54 validation, 54 test
Baseline	HOG + Logistic Regression
Improved model	MobileNetV3-Small transfer learning

Dataset and Data Collection

We collected the images from Wikimedia Commons with the official MediaWiki API (Wikimedia Commons contributors, n.d.). The source categories were People sitting, People standing, Female people waving hands, and Male people waving hands. This means the dataset was collected for this project and was not downloaded as a prepared machine-learning dataset. A fixed seed was used when selecting candidates, which makes the collection repeatable.

The script accepted JPEG, PNG, and WebP files, converted them to RGB JPEG, and limited the longest image side to 640 pixels. Images smaller than 180 pixels on either side were rejected. Exact duplicates were removed by SHA-256 and close duplicates were checked with a perceptual difference hash. The manifest records the original page, image URL, author or credit, license, dimensions, and checksum for every file. All 360 records have a source page and a license value.

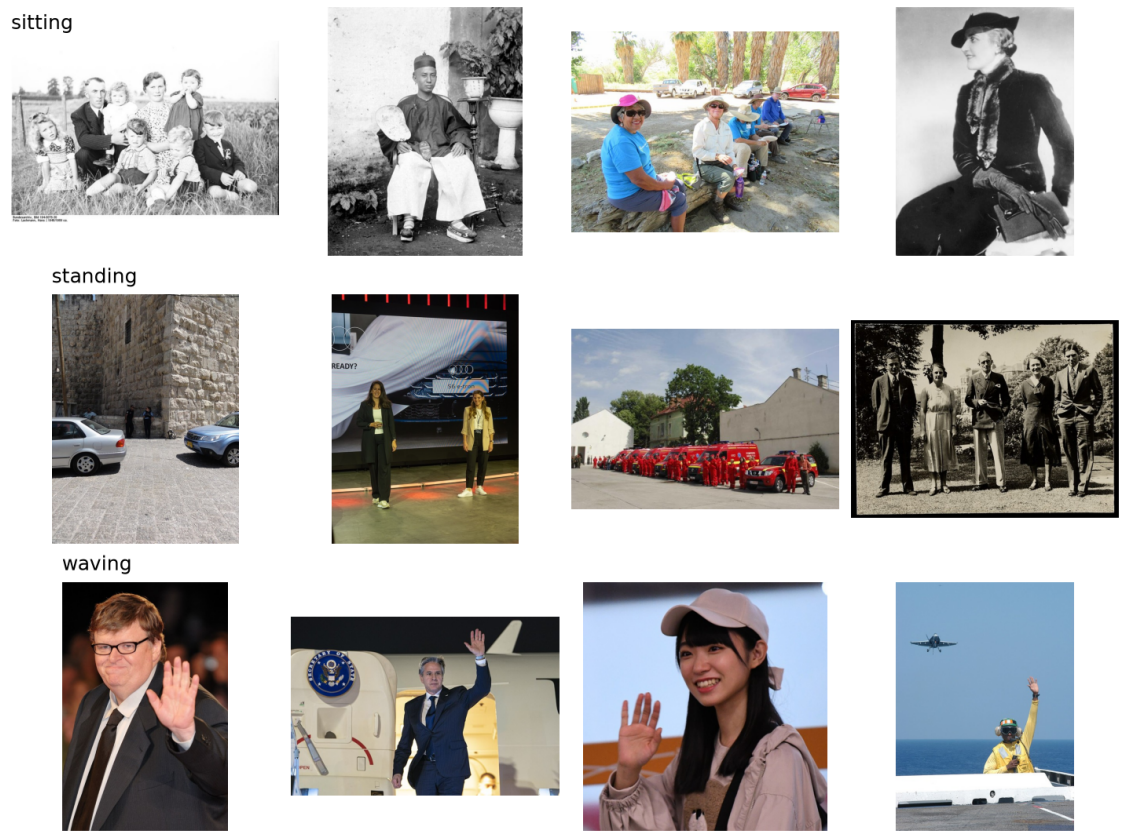


Figure 1. Example images from the three classes.

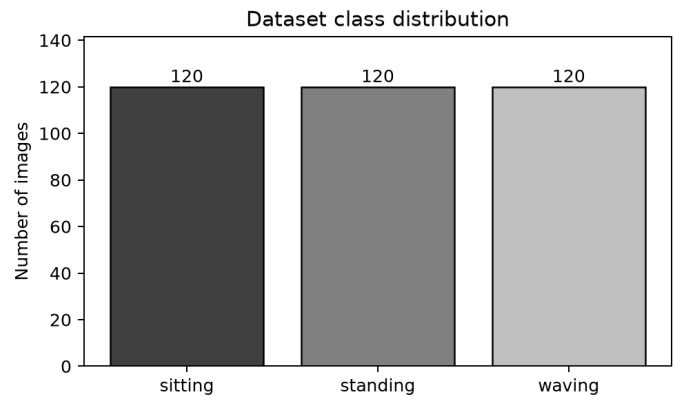


Figure 2. Balanced class distribution.

Preprocessing and Model Development

Data Split and Augmentation

We used a stratified 70/15/15 split with random seed 42. Each split contains the same number of examples from every class: 84 per class for training, 18 for validation, and 18 for testing. The test images were kept separate until the final evaluation.

For the baseline, images were resized to 128 x 128 and converted to grayscale. For the CNN, training images used random resized crop, horizontal flip, rotation up to 8 degrees, and small brightness and contrast changes. These operations create slightly different examples during training and reduce overfitting. Validation and test images used resize to 256, center crop to 224 x 224, conversion to tensor, and ImageNet normalization.

Baseline Model

The baseline uses Histograms of Oriented Gradients (HOG), a descriptor introduced for human detection by Dalal and Triggs (2005). HOG describes local edge directions and is useful for human shape. The HOG vectors were standardized and given to Logistic Regression. This model is fast and gives a clear reference point, but it cannot learn high-level visual features from the data.

Transfer-Learning Model

The improved model is MobileNetV3-Small, an efficient CNN architecture developed by Howard et al. (2019), with ImageNet pretrained weights from torchvision (PyTorch, n.d.). We froze the convolutional feature extractor and replaced the last layer with a three-class linear layer. The classifier was trained for 8 epochs with AdamW, learning rate 0.001, weight decay 0.0001, batch size 16, and cross-entropy loss. The checkpoint with the highest validation accuracy was used for testing. Training ran on Apple MPS acceleration.

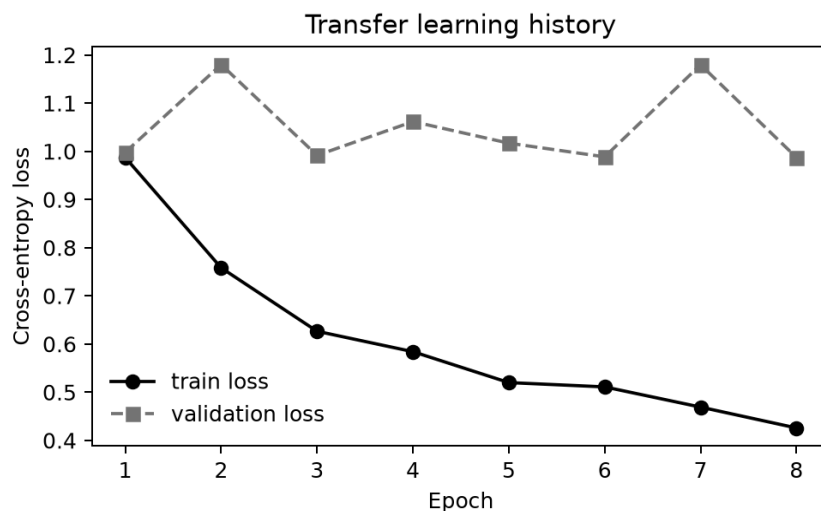


Figure 3. Training history. Best validation accuracy was 61.1%.

Results

The improved model performed much better on every main metric. Its accuracy was 77.8%, compared with 38.9% for the baseline. Macro F1 is important here because it gives equal weight to all three classes. The transfer model improved macro F1 from 0.389 to 0.779.

Model	Accuracy	Precision	Recall	F1
HOG + Logistic Regression	0.389	0.394	0.389	0.389
MobileNetV3-Small	0.778	0.783	0.778	0.779

MobileNetV3-Small results by class

Class	Precision	Recall	F1	Test images
sitting	0.875	0.778	0.824	18
standing	0.684	0.722	0.703	18
waving	0.789	0.833	0.811	18

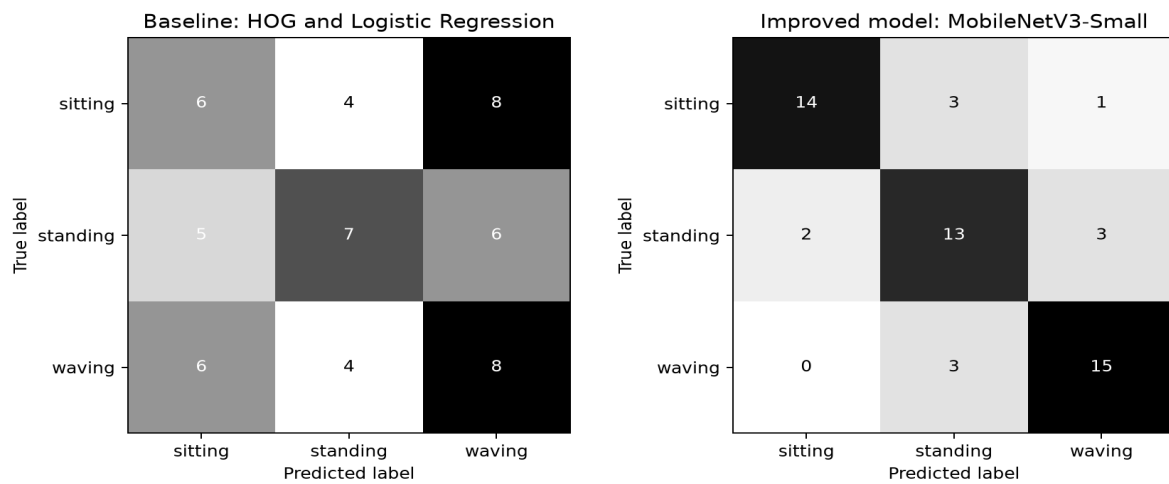


Figure 4. Confusion matrices for both models.

Discussion

The baseline was only slightly above the 33.3% random level. Its confusion matrix shows that it mixed all three classes. HOG mainly describes edges, so it can react to background structure, clothing, or image style instead of understanding the full pose. The transfer model correctly classified 42 of 54 test images. It performed best on sitting by precision and on waving by recall. Standing was the hardest class, with three examples predicted as waving and two as sitting.

The validation accuracy changed between epochs while training loss continued to go down. This suggests some overfitting, which is expected with only 252 training images. Saving the best validation checkpoint helped avoid using the last epoch automatically. The large improvement on the test set still shows the value of transfer learning.

Limitations and Future Work

The dataset was collected from public internet categories, so the labels were not checked by several independent annotators. Some images contain several people, a small person, or more than one possible action. The collection also includes different periods, cameras, and image styles. These factors make the task realistic but add label noise. The test set has only 18 images per class, so a few mistakes change the score.

A stronger next version would manually review all labels, crop the main person, collect more modern photos, and group related images before splitting to reduce source similarity. It would also be useful to fine-tune the last MobileNet feature blocks with a smaller learning rate and use cross-validation. For waving, short video clips could provide motion information that is missing from one photo.

Conclusion

This project completed the required data collection, preprocessing, augmentation, model comparison, and evaluation steps. A balanced 360-image dataset was built without Kaggle. MobileNetV3-Small reached 77.8% test accuracy and clearly outperformed the HOG baseline. The main lesson is that pretrained CNN features can learn useful pose information even when the custom dataset is small and visually varied.

References

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